

Ling Xu

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EDUCATION

University of British Columbia

September 2025 - Expected April 2029

Bachelor of Applied Science – Electrical Engineering

- Co-op Availability: 4, 8, or 12 months
- Relevant Courses: Electricity and Magnetism, Circuit Analysis, Differential Equations

TECHNICAL SKILLS

Hardware: STM32, Arduino, Magnetic Encoders, Stepper motors, Breadboarding & circuit debugging

Equipment: Oscilloscopes, Waveform generators, Multimeters, Soldering

Software & Signal Processing: C/C++, Python, MATLAB, Autocad, Mobile App Development, Cascade PID, I2C, Signal Filtering, State Feedback, UART

Certifications: Class 5 Alberta license, WHMIS certification

TECHNICAL PROJECTS

Inverted Pendulum Control System

July 2026 - September 2026

- Designed and built a self-balancing inverted pendulum from scratch to achieve real-time angular stability using an Arduino microcontroller and magnetic encoders.
- Implemented a dual-loop cascaded PID algorithm to precisely control stepper motor acceleration and maintain equilibrium via microstepping drivers.
- Eliminated encoder read errors caused by signal noise by calculating optimal I2C pull-up resistance and applying noise-mitigation wiring, maintaining high signal integrity at a 400 kHz bus speed.

Parallel-Oriented Tracking Device

May 2026 - July 2026

- Engineered a motorized positioning system to dynamically align and track mobile phone spatial orientation in real time using sensor fusion algorithms.
- Developed a custom mobile application to stream live orientation data at 10 Hz over Bluetooth/RF utilizing complementary filtering techniques.
- Corrected mechanical slippage by integrating an I2C encoder with a proportional state-feedback control loop, ensuring accurate long-term positioning tracking.

WORK EXPERIENCE

Calgary Stampede | Calgary, Alberta

July 2025, July 2026

Venue Services Attendant

- Maintained site sanitation and venue readiness across high-traffic facilities following WHMIS safety procedures.
- Coordinated with team members and independently executed tasks to optimize coverage efficiency.

VOLUNTEER EXPERIENCE

Wing Kei | Calgary, Alberta

January 2025 - June 2025

Volunteer Care Assistant

- Supported healthcare staff in delivering daily meal plans and personal care to ensure nutritional and health needs were met for 50+ senior residents.
- Executed mobility transfers and patient assistance routines to maintain a safe, patient-centered environment while adhering to mobility management guidelines.

AWARDS

Suncor Energy Scholarship | Suncor

2025

Alexander Rutherford Scholarship | Government of Alberta

2025

- Merit-based scholarship recognizing high academic achievement

INTERESTS & ACTIVITIES

- Biking and exploring
- 3D printing/prototyping & CAD modelling